

Tutorial on Number system

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Tutorial title: Introduction to python and Number system conversion

CO Mapping

Name	CO1	CO2	CO3
Number System Conversion	✓	-	-

In computer applications the **Binary numbers** are represented by only two symbols or digits, i.e. 0 and 1. The binary numbers here are expressed in the base-2 numeral system. For example, $(101)_2$ is a binary number. Each digit in this system is said to be a bit. The **Octal number system** has base value equal to 8 and comprise numbers 0 to 7. The **Decimal number system** is the number system that we use on a daily basis based on the 10 digits (0-9). The **Hexadecimal number system** number system has a base value equal to 16. containing numbers from 0 to 9 and the capital letters A to F to represent its Binary or Decimal number equivalent.

Conversion of Binary to Other Number Systems

To convert a **binary number to octal number** then Group all the 1's and 0's in the binary number in sets of three, starting from the far right. Add zeros to the left of the last digit if you don't have enough digits to make a set of three. Add up the new numbers in each set of three. For example: 10101_2 is the binary number. We can write the given binary number as: 010 101. Now as we know, in the octal number system, $010 \rightarrow 2$ and $101 \rightarrow 5$. Therefore, the required octal number is $(25)_8$.

To convert the **binary number to decimal** is equal to the sum of binary digits (dn) times their power of 2 (2^n). For example: $(10101)_2 = 1 \times 2^4 + 0 \times 2^3 + 1 \times 2^2 + 0 \times 2^1 + 1 \times 2^0 = 16 + 4 + 1 = (21)_{10}$.

To convert **binary number to hexadecimal**: Divide the binary digits into groups of four (starting from right) for integer part and start from left for fraction part. Convert each group of four binary digits to one hexadecimal digit. For example: $0001\ 0101 = (15)_{16}$.

Conversion of Octal to Other Number Systems

To convert the **octal to binary**: In order to convert the octal number into binary, we need to express every octal value using 3 binary bits. For example $(41)_8 = 100\ 001 = (100001)_2$. To convert the **octal to decimal**: is equal to the sum of octal digits (dn) times their power of 8 (8^n). For example: $41_8 = (4 \times 8^1) + (1 \times 8^0) = 4 \times 8 + 1 \times 1 = 32 + 1 = (33)_{10}$.

To convert the **octal to hexadecimal**: In order to convert the octal number into binary, we need to express every octal value using 3 binary bits. And after that, we need to group every 4 binary bits and calculate the value [From left to right]. For example: $(41)_8 = 100\ 001$. Combine the groups = $(100001)_2$. Divide the binary digits from previous groups of 4 digits, starting from the right. = $0010\ 0001$. Convert each group of 4 binary digits into 1 hexadecimal digit. $0010 = 2$, $0001 = 1$, = $(21)_{16}$.

Tutorial on Number system

Conversion of Decimal to Other Number Systems

To convert a **decimal number to binary** we need to divide the given number by 2 until the quotient is equal to 0. We keep the remainders aside during the division process. Once the quotient is equal to zero, we write the remainder along with the last numbers starting from the bottom to the top to obtain the binary number. For example, $(20)_{10} = (10000000)_2$

To convert **decimal to octal**, the decimal number is divided by 8 and record the remainders. Once the quotient is less than 8, we obtain the octal number by writing the remainder in reverse order. For example, $(45)_{10} = (55)_8$

The **decimal to hexadecimal** conversion is done similarly to the other two number systems. The base number of hexadecimal is 16 so the number needs to be divided by 16 until the quotient is zero.

Conversion of Hexadecimal to Other Number Systems

To convert **hexadecimal to a binary** number we need to first convert the hexadecimal number to a decimal number to finally convert it to a binary number. For example, $(100)_{16}$ is converted to decimal by multiplying each digit with 16^{n-1} .

$$(100)_{16} = 1 \times 16^{(3-1)} + 0 \times 16^{(2-1)} + 0 \times 16^{(1-1)}$$

$$(100)_{16} = 1 \times 16^2 + 0 \times 16^1 + 0 \times 16^0$$

$$(100)_{16} = (256)_{10}$$

Now, convert the decimal number $(256)_{10}$ to a binary number by dividing the number by 2 until the quotient is zero.

$$(100)_{16} = (100000000)_2$$

Questions:

1. Which type of Programming does Python support?
 - a) object-oriented programming
 - b) structured programming
 - c) functional programming
 - d) All of the mentioned
2. Which of the following is the correct extension of the Python file?
 - a) .python
 - b) .pl
 - c) .py
 - d) .p
3. Convert the following numbers from base 10 to base 8-
 - a. $(1032)_{10}$
 - b. $(1032.6875)_{10}$
 - c. $(172)_{10}$
 - d. $(172.878)_{10}$

Tutorial on Number system

4. Convert the following numbers from base 10 to base 2-
- $(18)_{10}$
 - $(18.625)_{10}$
 - $(172)_{10}$
 - $(172.878)_{10}$
5. Convert the following numbers to base 10-
- $(10010)_2$
 - $(254)_8$
 - $(10010.101)_2$
 - $(254.7014)_8$
 - $(0.1402)_8$
6. Convert Hexadecimal to Octal : $(1056)_{16}$ to $(?)_8$
7. Convert Octal to Hexadecimal : $(11672)_8$ to $(?)_{16}$
8. Convert Octal to any base like 5: $(2724)_8$ to $(?)_5$
9. Convert Binary to any base like 6: $(1001001100)_2$ to $(?)_6$
10. Convert the following numbers to base 10-
- $(AC)_{16}$
 - $(AC.FBA5)_{16}$
 - $(0.ABDF)_{16}$